

XING SHENG

Professor

xingsheng@tsinghua.edu.cn

<https://shengxingstars.github.io/www/>

Department of Electronic Engineering
Rohm Building, Tsinghua University
Beijing, China 100084

Education

- *Ph.D., Materials Science and Engineering, Massachusetts Institute of Technology, 2012*
Advisor: Lionel C. Kimerling
- *B. Eng., Materials Science and Engineering, Tsinghua University, 2007*

Professional Experiences

- *Endowed Associate Professor, Dept. Electronic Engineering, Tsinghua University, present*
also affiliated with: IDG/McGovern Institute for Brain Research at Tsinghua University
- *Postdoctoral Associate, University of Illinois at Urbana-Champaign, 2012–2015*
Advisor: John A. Rogers

Research Interests

- Non-conventional Optoelectronics for Biomedical Applications
- Optical Neural Interfaces
- Biocompatible and Biodegradable Photonics

Teaching Experience

- Leading Lecturer at Tsinghua
 - 20230313 “Foundation of Solid State Physics”
 - 80230992 “Principles of Micro- and Nanofabrication for Electronic and Photonic Devices”
 - 80231001 “Laboratory of Micro- and Nanofabrication for Electronic and Photonic Devices”
 - 60230072 “Academic Writings and Presentations for Electrical Engineering”
- Worked as a guest lecturer and a teaching assistant for multiple courses at Tsinghua, MIT and UIUC
- Supervised > 20 undergraduate and > 10 graduate students at Tsinghua, MIT and UIUC

Lecture notes for these courses have been posted online and received widespread attention:

<https://shengxingstars.github.io/www/teaching.html>

Publications

Peer-Reviewed Journals:

Google Scholar: <https://scholar.google.com/citations?hl=en&user=bS9skH4AAAAJ>

#co-first author, *corresponding author

1. Y. Xie, Y. Peng, J. Guo, M. Liu, B. Zhang, L. Yin, H. Ding*, **X. Sheng***, “Materials and Devices for High-Density, High-Throughput Micro-Electrocorticography Arrays”, *Fundamental Research* **XX**, XX–XX (2025) (**Front Cover**).
2. **X. Sheng***, Y. Sui, L. Li, J. A. Rogers, “Multimodal technologies for neural modulation and sensing”, *Fundamental Research* **XX**, XX–XX (2025).
3. X. Cai#, H. Zhang#, P. Wei#, Q. Liu#, D. Sheng, Z. Li, B. Zhang, G. Tang, W. Zhao, Z. Ye, Z. Xue, Y. Xie, Y. Dai, C. Wang, Y. Wang, X. Fu, L. Yin, H. Peng, H. Ding*, G. Zhao*, **X. Sheng***, “A wireless optoelectronic probe to monitor oxygenation in deep brain tissue”, *Nature Photonics* **18**, 492–500 (2024) (**Front Cover**).
4. X. Fu#, Z. Hu#, W. Li, L. Ma, J. Chen, M. Liu, J. Liu, S. Hu, H. Wang, Y. Huang, G. Tang, B. Zhang, X. Cai, Y. Wang, L. Li, J. Ma, S.-H. Shi, L. Yin, H. Zhang, X. Li*, **X. Sheng***, “A Silicon Diode-based Optoelectronic Interface for Bidirectional Neural Modulation”, *Proceedings of the National Academy of Sciences USA* **121**, e2404164121 (2024).
5. J. Chen, H. Ding*, Y. Li, **X. Sheng***, “Numerical Design of a Micro-LED based Optogenetic Stimulator for Visual Cortical Prosthesis”, *IEEE Journal of Selected Topics in Quantum Electronics* **30**, 7200308 (2024) (**Front Cover**).
6. J. Chen, H. Ding*, **X. Sheng***, “Advanced Manufacturing of Microscale Light-Emitting Diodes and Their Use in Displays and Biomedicine”, *Journal of Information Display* **25**, 1–12 (2024) (**Invited**).
7. Q. Du, H. Dai, Y. Sui, **X. Sheng***, “Recent Progress on Multimodal Technologies for Neural Modulation and Sensing”, *Bulletin of National Natural Science Foundation of China* **38**, 296–302 (2024) (**Front Cover**).
杜全生, 戴慧, 眭亚楠, **盛兴***, 多模态神经调控与检测技术研究进展, 中国科学基金, **38**, 296–302 (2024).
8. L. Li, B. Zhang, W. Zhao, D. Sheng, L. Yin, **X. Sheng***, D. Yao*, “Multimodal Technologies for Closed-Loop Neural Modulation and Sensing”, *Advanced Healthcare Materials* **13**, 2303289 (2024) (**Invited**).
9. Z. Xue, W. Chou, Y. Xu, Z. Cheng, X. Ren, T. Sun, W. Tong, Y. Xie, J. Chen, N. Zhang, **X. Sheng**, Y. Wang, H. Zhao, J. Yang*, H. Ding*, “Battery-free Optoelectronic Patch for Photodynamic and Light Therapies in Treating Bacteria-infected Wounds”, *Biosensors and Bioelectronics* **261**, 116467 (2024).
10. B. Yu, J. Bai, Y. Guan, X. Huang, L. Liang, Z. Ren, X. Song, T. Zhang, C. Yang, F. Dai, X. Wang, **X. Sheng**, J. Peng, L. Wang*, Y. Wang*, L. Yin*, “Fully biodegradable and self-powered nerve guidance conduit based on zinc-molybdenum batteries for peripheral nerve repair”, *Biosensors and Bioelectronics* **263**, 116578 (2024).
11. L. Wang, S. Liu, W. Zhao, J. Li, H. Zeng, S. Kang, **X. Sheng**, L. Wang*, Y. Fan*, L. Yin*, “Recent Advances in Implantable Neural Interfaces for Multimodal Electrical Neuromodulation”, *Advanced Healthcare Materials* **13**, 2303316 (2024) (**Invited**).
12. C. Hu, L. Wang, S. Liu, **X. Sheng**, L. Yin*, “Recent Development of Implantable Chemical Sensors Utilizing Flexible and Biodegradable Materials for Biomedical Applications”, *ACS Nano* **18**, 3969–3995 (2024).

13. P. Sun, C. Li, C. Yang, M. Sun, H. Hou, Y. Guan, J. Chen, S. Liu, K. Chen, Y. Ma, Y. Huang, X. Li, H. Wang, L. Wang, S. Chen, H. Cheng, W. Xiong, **X. Sheng**, M. Zhang, J. Peng, S. Wang*, Y. Wang*, L. Yin*, “A Biodegradable and Flexible Neural Interface for Transdermal Optoelectronic Modulation and Regeneration of Peripheral Nerves”, *Nature Communications* **15**, 4721 (2024).
14. X. Cheng, W. Li, Y. Wang, K. Weng*, Y. Xing, Y. Huang, **X. Sheng**, J. Yao*, H. Zhang*, J. Li*, “Highly Branched Au Superparticles as Efficient Photothermal Transducers for Optical Neuromodulation”, *ACS Nano* **18**, 29572–29584 (2024).
15. X. Wang, A. Aziz, **X. Sheng**, L. Wang*, L. Yin*, “Bioresorbable neural interfaces for bioelectronic medicine”, *Current Opinion in Biomedical Engineering* **32**, 100565 (2024).
16. J. Liu, Z. Li, M. Sun, L. Zhou, X. Wu, Y. Lu, Y. Shao, C. Liu, N. Huang, B. Hu, Z. Wu, C. You, L. Li, M. Wang, L. Tao, Z. Di, **X. Sheng**, Y. Mei*, E. Song*, “Flexible Bioelectronic Systems with Large-Scale Temperature Sensor Array for Monitoring and Treatments of Localized Wound Inflammation”, *Proceedings of the National Academy of Sciences USA* **121**, e2412423121 (2024).
17. Y. M. Song, Y. Chai, D.-H. Kim, **X. Sheng**, D. Son, S. Kim, “Editorial Interview Flexible Optoelectronics”, *IEEE Journal of Selected Topics in Quantum Electronics* **30**, 1000403 (2024).
18. Y. Huang#, Y. Cui#, H. Deng#, J. Wang, R. Hong, S. Hu, H. Hou, Y. Dong, H. Wang, J. Chen, L. Li, Y. Xie, P. Sun, X. Fu, L. Yin, W. Xiong, S.-H. Shi, M. Luo, S. Wang*, X. Li*, **X. Sheng***, “Bioresorbable Thin-Film Silicon Diodes for the Optoelectronic Excitation and Inhibition of Neural Activities”, *Nature Biomedical Engineering* **7**, 486–498 (2023) (**Front Cover**).
19. H. Wang, J. Tian, Y. Jiang, S. Liu, J. Zheng, N. Li, G. Wang, F. Dong, J. Chen, Y. Xie, Y. Huang, X. Cai, X. Wang, W. Xiong, H. Qi, L. Yin, Y. Wang*, **X. Sheng***, “A 3D Biomimetic Optoelectronic Scaffold Repairs Cranial Defects”, *Science Advances* **9**, abq7750 (2023).
20. H. Ding*, Y. Peng, G. Lv, Y. Xie, J. Chen, Z. Shi, Y. Deng, L. Yin, J. Yang, Y. Wang, **X. Sheng***, “Heterogeneous Integration of Thin-Film Organic and Inorganic Devices for Optical based Bioelectrical and Chemical Sensing”, *IEEE Journal of Selected Topics in Quantum Electronics* **29**, 5200107 (2023) (**Invited**).
21. **X. Sheng***, W. Zhao, L. Li, Y. Huang, H. Ding, “Foundation of Brain-Machine Interfaces: Neurons and Diodes”, *Chinese Journal of Lasers* **50**, 0907301 (2023) (**Invited**) (**Front Cover**).
 盛兴*, 赵汶鑫, 李丽珠, 黄云翔, 丁贺, 脑机接口技术的基础研究: 神经元与二极管, 中国激光, **50**, 0907301 (2023) (**Front Cover**). *生物医学光子学-最美封面论文*
22. Z. Shi#, X. Gao#, X. Cai, H. Zhao, X. Wang, L. Zhao, L. Yin, H. Ding*, **X. Sheng***, “Fully Implantable and Retrievable Upconversion Waveguides for Photodynamic Therapy in Deep Tissue”, *Advanced Optical Materials* **11**, 2300689 (2023) (**Inside Front Cover**).
23. K. Zhang, W. Zhao, **X. Sheng***, “Ion-Gated Tungsten Oxide Based Electrochemical Transistors with Subthreshold Slopes Approaching the Thermodynamic Limit”, *Applied Physics A* **129**, 728 (2023).
24. Y. Cui#, L. Li#, C. Liu*, Y. Wang, M. Sun, B. Jia, Z. Shen, **X. Sheng***, Y. Deng*, “Water-Responsive 3D Electronics for Smart Biological Interfaces”, *Nano Letters* **23**, 11693–11701 (2023).
25. P. Sun, Y. Guan, C. Yang, H. Hou, S. Liu, B. Yang, X. Li, S. Chen, L. Wang, H. Wang, Y. Huang, **X. Sheng**, J. Peng, W. Xiong, Y. Wang*, L. Yin*, “A Bioresorbable and Conductive Scaffold Integrating Silicon Membranes for Peripheral Nerve Regeneration”, *Advanced Healthcare Materials* **12**, 2301859 (2023).
26. S. Liu, Q. Yu, R. Guo, K. Chen, J. Xia, Z. Guo, L. He, Q. Wu, L. Liu, Y. Li, B. Zhang, L. Lu, **X. Sheng**, J. Zhu, L. Zhao, H. Qi, K. Liu*, L. Yin*, “A Biodegradable, Adhesive and Stretchable Hydrogel and Potential Applications for Allergic Rhinitis and Epistaxis”, *Advanced Healthcare Materials* **12**, 2302059 (2023).
27. L. Kong, H. Wen, Y. Luo, X. Chen, **X. Sheng**, Y. Liu*, P. Chen*, “Dual-Conductive and Stiffness-Morphing Microneedle Patch Enables Continuous in Planta Monitoring of Electrophysiological Signal

- and Ion Fluctuation”, *ACS Applied Materials & Interfaces* **15**, 43515–43523 (2023).
28. X. Huang, H. Hou, B. Yu, J. Bai, Y. Guan, L. Wang, K. Chen, X. Wang, P. Sun, Y. Deng, S. Liu, X. Cai, Y. Wang, J. Peng, **X. Sheng**, W. Xiong*, L. Yin*, “Fully Biodegradable and Long-Term Operational Primary Zinc Batteries as Power Sources for Electronic Medicine”, *ACS Nano* **17**, 5727–5739 (2023).
 29. Y. Deng, M. Zhao, Y. Ma, S. Liu, M. Liu, B. Shen, R. Li, H. Ding, H. Cheng, **X. Sheng**, W. Fu, Z. Li, M. Zhang, L. Yin*, “A Flexible and Biomimetic Olfactory Synapse with Gasotransmitter-Mediated Plasticity”, *Advanced Functional Materials* **33**, 2214139 (2023) (**Back Cover**).
 30. F. Dai, Q. Geng, T. Hua, **X. Sheng**, L. Yin*, “Organic Biodegradable Piezoelectric Materials and Their Potential Applications as Bioelectronics”, *Soft Science* **3**, 7 (2023) (**Invited**).
 31. S. Liu, X. Wang, S. Liu, Y. Deng, B. Zhao, H. Wang, **X. Sheng**, L. Zhao, L. Wang*, P. Zhang*, L. Yin*, “Laser-Triggered Degradation of Silicon Circuits by Lithiation and Moisture Uptake for On-Demand Transient Electronics”, *Advanced Engineering Materials* **25**, 2300213 (2023).
 32. Q. Zhou, X. Fu, J. Xu, S. Dong, C. Liu, D. Cheng, C. Gao, M. Huang, Z. Liu, X. Ni, R. Hua, H. Tu, H. Sun, Q. Shen, B. Chen, J. Zhang, L. Zhang, H. Yang, J. Hu, W. Yang, W. Pei, Q. Yao, **X. Sheng**, J. Zhang*, W.Z. Yang*, W.L. Shen*, “Hypothalamic Warm-Sensitive Neurons Require TRPC4 Channel for Detecting Internal Warmth and Regulating Body Temperature in Mice”, *Neuron* **111**, 387–404 (2023).
 33. Y. Luo, X. Chen*, *et al.*, “Technology Roadmap for Flexible Sensors”, *ACS Nano* **17**, 5211–5295 (2023).
 34. L. Li#, L. Lu#, Y. Ren#, G. Tang#, Y. Zhao, X. Cai, Z. Shi, H. Ding, C. Liu, D. Cheng, Y. Xie, H. Wang, X. Fu, L. Yin, M. Luo*, **X. Sheng***, “Colocalized, Bidirectional Optogenetic Modulations in Freely Behaving Mice with a Wireless Dual-Color Optoelectronic Probe”, *Nature Communications* **13**, 839 (2022).
 35. X. Cai#, L. Li#, W. Liu, N. Du, Y. Zhao, Y. Han, C. Liu, Y. Yin, X. Fu, D. Sheng, L. Yin, L. Wang, P. Wei*, **X. Sheng***, “A Dual-Channel Optogenetic Stimulator Selectively Modulates Distinct Defensive Behaviors”, *iScience* **25**, 103681 (2022) (**Invited**).
 36. H. Ding*, G. Lv, X. Cai, J. Chen, Z. Cheng, Y. Peng, G. Tang, Z. Shi, Y. Xie, X. Fu, L. Yin, J. Yang, Y. Wang, **X. Sheng***, “An Optoelectronic Thermometer based on Microscale Infrared-to-Visible Conversion Devices”, *Light: Science & Applications* **11**, 130 (2022).
 37. H. Wang, J. Tian, B. Lu, Y. Xie, P. Sun, L. Yin, Y. Wang, **X. Sheng***, “Degradation Study of Thin-Film Silicon Structures in a Cell Culture Medium”, *Sensors* **22**, 802 (2022) (**Invited**).
 38. R. Nazempour#, B. Zhang#, Z. Ye, L. Yin, X. Lv, **X. Sheng***, “Emerging Applications of Optical Fiber-Based Devices for Brain Research”, *Advanced Fiber Materials* **4**, 24–42 (2022) (**Invited Review**).
 39. D. Kong, Y. Zhang, D. Cheng, E. Wang, K. Zhang, H. Wang, K. Liu, L. Yin*, **X. Sheng***, “Heteroepitaxy of Large-Area, Monocrystalline Lead Halide Perovskite Films on Gallium Arsenide”, *ACS Applied Materials & Interfaces* **14**, 52508–52515 (2022).
 40. D. Kong#, K. Zhang#, J. Tian, L. Yin*, **X. Sheng***, “Biocompatible and Biodegradable Light-Emitting Materials and Devices”, *Advanced Materials Technologies* **7**, 2100006 (2022) (**Invited Review**).
 41. Y. Deng, H. Qi, Y. Ma, S. Liu, M. Zhao, Z. Guo, Y. Jie, R. Zheng, J. Jing, K. Chen, H. Ding, G. Lv, K. Zhang, R. Li, H. Cheng, L. Zhao, **X. Sheng**, M. Zhang*, L. Yin*, “A flexible and highly sensitive organic electrochemical transistor-based biosensor for continuous and wireless nitric oxide detection”, *Proceedings of the National Academy of Sciences USA* **119**, e2208060119 (2022).
 42. S. Lu, Z. Fu, F. Li, K. Weng, L. Zhou, L. Zhang, Y. Yang, H. Qiu, D. Liu, W. Qing, H. Ding, **X. Sheng**, M. Chen, X. Tang, L. Duan, W. Liu, L. Wu, Y. Yang, H. Zhang*, J. Li, “Beyond a Linker: The Role of Photochemistry of Crosslinkers in the Direct Optical Patterning of Colloidal Nanocrystals”,

Angewandte Chemie **134**, e202202633 (2022).

43. L. Li, G. Tang, Z. Shi, H. Ding, C. Liu, D. Cheng, Q. Zhang, L. Yin, Z. Yao, L. Duan, D. Zhang, C. Wang, M. Feng, Q. Sun, Q. Wang, Y. Han, L. Wang, Y. Luo, **X. Sheng***, “Transfer-Printed, Tandem Microscale Light-Emitting Diodes for Full-Color Displays”, *Proceedings of the National Academy of Sciences USA* **118**, e2023436118 (2021).
44. H. Ding, G. Lv, Z. Shi, D. Cheng, Y. Xie, Y. Huang, L. Yin, J. Yang, Y. Wang, **X. Sheng***, “Optoelectronic Sensing of Biophysical and Biochemical Signals based on Photon Recycling of a micro-LED”, *Nano Research* **14**, 3208–3213 (2021) (*Invited*) (*Front Cover*).
45. Y. Xie, H. Wang, D. Cheng, H. Ding, D. Kong, L. Li, L. Yin, G. Zhao, L. Liu, G. Zou, J. Wei, C. Li, C. Liu*, **X. Sheng***, “Diamond Thin Films Integrated with Flexible Substrates and Their Physical, Chemical and Biological Characteristics”, *Journal of Physics D: Applied Physics* **54**, 384004 (2021) (*Invited*).
46. Y. Yang, M. Wu, A. Vázquez-Guardado, A. Wegener, J. Grajales-Reyes, Y. Deng, T. Wang, R. Avila, J. Moreno, S. Minkowicz, V. Dumrongprechachan, J. Lee, S. Zhang, A. Legaria, Y. Ma, S. Mehta, D. Franklin, L. Hartman, W. Bai, M. Han, H. Zhao, W. Lu, Y. Yu, **X. Sheng**, A. Banks, X. Yu, Z. Donaldson, R. Gereau, C. Good, Z. Xie*, Y. Huang*, Y. Kozorovitskiy*, J. A Rogers*, “Wireless multilateral devices for optogenetic studies of individual and social behaviors”, *Nature Neuroscience* **24**, 1035–1045 (2021).
47. C. Liu, Y. Zhao, X. Cai, Y. Xie, T. Wang, D. Cheng, L. Li, R. Li, Y. Deng, H. Ding, G. Lv, G. Zhao, L. Liu, G. Zou, M. Feng, Q. Sun, L. Yin, **X. Sheng***, “A Wireless, Implantable Optoelectrochemical Probe for Optogenetic Stimulation and Dopamine Detection”, *Microsystems & Nanoengineering* **6**, 64 (2020) (*Invited*) (*Front Cover*).
48. R. Nazempour, Q. Zhang, C. Liu, **X. Sheng***, “Design of Silicon Photonic Structures for Multi-site, Multi-Spectral Optogenetics in the Deep Brain”, *IEEE Photonics Journal* **12**, 4200107 (2020).
49. D. Kong#, D. Cheng#, X. Wang, K. Zhang, H. Wang, K. Liu, H. Li, **X. Sheng***, L. Yin*, “Solution Processed Lead-free Cesium Titanium Halide Perovskites and Their Structural, Thermal and Optical Characteristics”, *Journal of Materials Chemistry C* **8**, 1591–1597 (2020) (*Inside Front Cover*).
50. Y. Jiang, W. Qi, Q. Zhang, H. Liu, J. Zhang, N. Du, R. Nazempour, Y. Su, R. Fu, K. Zhang, P. Lyu, F. Dong, L. Yin, **X. Sheng***, Y. Wang*, “Green Light-Based Photobiomodulation with an Implantable and Biodegradable Fiber for Bone Regeneration”, *Small Methods* **4**, 1900879 (2020) (*Back Cover*).
51. H. Wang#, P. Sun#, L. Yin*, **X. Sheng***, “3D Electronic and Photonic Structures as Active Biological Interfaces”, *InfoMat* **2**, 527–552 (2020) (*Invited Review*).
52. **X. Sheng***, L. Gao, Y. M. Song, H. Tao, S. Yun, “Bio-Inspired and Bio-Integrated Photonic Materials and Devices: Feature Issue Introduction”, *Optical Materials Express* **10**, 155–156 (2020) (*Invited*).
53. R. Li, H. Qi, Y. Ma, Y. Deng, S. Liu, Y. Jie, J. Jing, J. He, X. Zhang, L. Wheatley, C. Huang, **X. Sheng**, M. Zhang, L. Yin*, “A Flexible and Physically Transient Electrochemical Sensor for Real-Time Wireless Nitric Oxide Monitoring”, *Nature Communications* **11**, 3207 (2020).
54. L. Wang, C. Lu, S. Yang, P. Sun, Y. Wang*, Y. Guan, S. Liu, D. Cheng, H. Meng, Q. Wang, J. He, H. Hou, H. Li, W. Lu, Y. Zhao, J. Wang, Y. Zhu, Y. Li, D. Luo, T. Li, H. Chen, S. Wang, **X. Sheng**, W. Xiong, X. Wang, J. Peng*, L. Yin*, “A fully biodegradable and self-electrified device for neuroregenerative medicine”, *Science Advances* **6**, eabc6686 (2020).
55. C. Jia, L. Li, Y. Liu, B. Fang, H. Ding, J. Song, Y. Liu, K. Xiang, S. Lin, Z. Li, W. Si, B. Li, **X. Sheng**, D. Wang, X. Wei*, H. Wu*, “Highly Compressible and Anisotropic Lamellar Ceramic Sponges with Superior Thermal Insulation and Acoustic Absorption Performances”, *Nature Communications* **11**, 3732 (2020).
56. R. Zhou, M. Feng, J. Wang, Q. Sun*, J. Liu, S. Zhang, M. Ikeda, T. Liu, Z. Huang, **X. Sheng**, H. Yang, “InGaN-based lasers with an inverted ridge waveguide heterogeneously integrated on Si(100)”,

ACS Photonics **7**, 2636–2642 (2020).

57. J. Liu, H. Qie, Q. Sun*, M. Feng, J. Wang, X. Sun, **X. Sheng**, M. Ikeda, H. Yang, “Enhanced carrier confinement and radiative recombination in GaN-based laser by tailoring first-barrier doping”, *Optics Express* **28**, 32124–32131 (2020).
58. J. Wang, M. Feng*, R. Zhou, Q. Sun*, J. Liu, X. Sun, X. Zheng*, M. Ikeda, **X. Sheng**, H. Yang, “Continuous-wave electrically injected GaN-on-Si microdisk laser diodes”, *Optics Express* **28**, 12201–12208 (2020).
59. J. Wang, M. Feng*, R. Zhou, Q. Sun*, J. Liu, X. Sun, X. Zheng*, **X. Sheng**, H. Yang, “Thermal characterization of electrically injected GaN-based microdisk lasers on Si”, *Applied Physics Express* **13**, 074002 (2020).
60. Y. Tang, M. Feng*, P. Wen, J. Liu, J. Wang, X. Sun, Q. Sun*, S. Zhang*, **X. Sheng**, M. Ikeda, H. Yang, “Degradation study of InGaN-based laser diodes grown on Si”, *Journal of Physics D: Applied Physics* **53**, 395103 (2020).
61. W. Ling, J. Yu, N. Ma, Y. Li, Z. Wu, R. Liang, Y. Hao, H. Pan, W. Liu, B. Fu, K. Wang, H. Wang, L. Li, **X. Sheng**, H. Peng, B. Ning, J. Yang, X. Huang*, “Flexible Electronics and Materials for Synchronized Stimulation and Monitoring in Multi-Encephalic Regions”, *Advanced Functional Materials* **30**, 2002644 (2020).
62. X. Huang, L. Wang*, H. Wang, B. Zhang, X. Wang, R. Y. Z. Stening, **X. Sheng**, L. Yin*, “Materials Strategies and Device Architectures of Emerging Power Supply Devices for Implantable Bioelectronics”, *Small* **16**, 1902827 (2020) (*Invited Review*).
63. D. Cheng, D. Kong, **X. Sheng**, L. Yin, H. Li*, “Perovskite Hetero-Anionic-Sublattice Interfaces for Optoelectronics and Nonconventional Electronics”, *Nanoscale* **12**, 7263–7272 (2020).
64. H. Zhou, J. Park, Y. Lee, J. Park, J. Kim, J. Kim, H. Lee, S. Jo, X. Cai, L. Li, **X. Sheng**, H. Yun, J. Park, J. Sun, T. Lee*, “Water Passivation of Perovskite Nanocrystals Enables Air-Stable Intrinsically Stretchable Color-Conversion Layers for Stretchable Displays”, *Advanced Materials* **32**, 2001989 (2020).
65. Z. Shi#, H. Ding#, H. Hong, D. Cheng, K. Rajabi, J. Yang, Y. Wang, L. Wang, Y. Luo, K. Liu, **X. Sheng***, “Ultrafast and Low-Power Optoelectronic Infrared-to-Visible Upconversion Devices”, *Photonics Research* **7**, 1161–1168 (2019) (*Invited*).
66. H. Ding, H. Hong, D. Cheng, Z. Shi, K. Liu, **X. Sheng***, “Power- and Spectral-Dependent Photon-Recycling Effects in a Double-Junction Gallium Arsenide Photodiode”, *ACS Photonics* **6**, 59–65 (2019).
67. Y. Zhao#, C. Liu#, Z. Liu, W. Luo, L. Li, X. Cai, D. Liang, Y. Su, H. Ding, Q. Wang, L. Yin, J. Guan, M. Luo, **X. Sheng***, “Wirelessly Operated, Implantable Optoelectronic Probes for Optogenetics in Freely Moving Animals”, *IEEE Transactions on Electron Devices* **66**, 785–792 (2019).
68. R. Nazempour, C. Liu, Y. Chen, C. Ma, **X. Sheng***, “Performance Evaluation of an Implantable Sensor for Deep Brain Imaging: an Analytical Investigation”, *Optical Materials Express* **9**, 3729–3737 (2019).
69. S. Xing, L. Lin, J. Huo, G. Zou, **X. Sheng**, L. Liu*, Y. N. Zhou, “Plasmon-Induced Heterointerface Thinning for Schottky Barrier Modification of Core/Shell SiC/SiO₂ Nanowires”, *ACS Applied Materials & Interfaces* **11**, 9326–9332 (2019).
70. L. Wang, Y. Gao, F. Dai, D. Kong, H. Wang, P. Sun, Z. Shi, **X. Sheng**, B. Xu*, L. Yin*, “Geometrical and Chemical Dependent Hydrolysis Mechanisms of Silicon Nanomembranes for Biodegradable Electronics”, *ACS Applied Materials & Interfaces* **11**, 18013–18023 (2019).
71. Y. Chen, H. Wang, Y. Zhang, R. Li, C. Chen, H. Zhang, S. Tang, S. Liu, X. Chen, H. Wu, R. Lv, **X. Sheng**, P. Zhang, S. Wang, L. Yin*, “Electrochemically-Triggered Degradation of Silicon Membranes for Smart On-Demand Transient Electronic Devices”, *Nanotechnology* **30**, 394002 (2019) (*Invited*).

72. W. Bai, J. Shin, R. Fu, I. Kandela, D. Lu, X. Ni, Y. Park, Z. Liu, T. Hang, D. Wu, Y. Liu, C. Haney, I. Stepien, Q. Yang, J. Zhao, K. Nandoliya, H. Zhang, **X. Sheng**, L. Yin, K. MacRenaris, A. Brikha, F. Aird, M. Pezhouh, J. Hornick, W. Zhou, J. A. Rogers*, “Bioresorbable Photonic Devices for the Spectroscopic Characterization of Physiological Status and Neural Activity”, *Nature Biomedical Engineering* **3**, 644–654 (2019).
73. E. Song, C. Chiang, R. Li, X. Jin, J. Zhao, M. Hill, Y. Xia, L. Li, Y. Huang, S. M. Won, K. J. Yu, **X. Sheng**, H. Fang, M. A. Alam, Y. Huang, J. Viventi, J. Chang, J. A. Rogers*, “Flexible Electronic/Optoelectronic Microsystems with Scalable Designs for Chronic Bio-Integration”, *Proceedings of the National Academy of Sciences USA* **116**, 15398–15406 (2019).
74. H. Ding#, L. Lu#, Z. Shi, D. Wang, L. Li, X. Li, Y. Ren, C. Liu, D. Cheng, H. Kim, N. C. Giebink, X. Wang, L. Yin, L. Zhao, M. Luo, **X. Sheng***, “Microscale Optoelectronic Infrared-to-Visible Upconversion Devices and Their Use as Injectable Light Sources”, *Proceedings of the National Academy of Sciences USA* **115**, 6632–6637 (2018).
75. C. Liu#, Q. Zhang#, D. Wang, G. Zhao, X. Cai, L. Li, H. Ding, K. Zhang, H. Wang, D. Kong, L. Yin, L. Liu, G. Zou, L. Zhao, **X. Sheng***, “High Performance, Biocompatible Dielectric Thin-Film Optical Filters Integrated with Flexible Substrates and Microscale Optoelectronic Devices”, *Advanced Optical Materials* **6**, 1800146 (2018) (**Back Cover**).
76. R. Fu#, W. Luo#, R. Nazempour#, D. Tan, H. Ding, K. Zhang, L. Yin, J. Guan*, **X. Sheng***, “Implantable and Biodegradable Poly(L-lactic acid) Fibers for Optical Neural Interfaces”, *Advanced Optical Materials* **6**, 1700941 (2018).
77. L. Li#, C. Liu#, Y. Su, J. Bai, J. Wu, Y. Han, Y. Hou, S. Qi, Y. Zhao, H. Ding, Y. Yan, L. Yin, P. Wang, Y. Luo, **X. Sheng***, “Heterogeneous Integration of Microscale GaN Light Emitting Diodes and Their Electrical, Optical and Thermal Characteristics on Flexible Substrates”, *Advanced Materials Technologies* **3**, 1700239 (2018) (**Back Cover**).
78. L. Lu#, P. Gutruf#, L. Xia#, D. Bhatti, X. Wang, A. Vazquez-Guardado, X. Ning, X. Shen, T. Sang, R. Ma, G. Pakeltis, G. Sobczak, H. Zhang, D. Seo, M. Xue, L. Yin, D. Chanda, **X. Sheng***, M. Bruchas*, J. A. Rogers*, “Wireless Optoelectronic Photometers for Monitoring Neuronal Dynamics in the Deep Brain”, *Proceedings of the National Academy of Sciences USA* **115**, E1374–E1383 (2018).
79. L. Lu#, Z. Yang#, K. Meacham, C. Cvetkovic, E. A. Corbin, A. Vazquez-Guardado, M. Xue, L. Yin, J. Boroumand, G. Pakeltis, T. Sang, K. J. Yu, D. Chanda, R. Bashir, R. W. Gereau, **X. Sheng***, J. A. Rogers*, “Biodegradable Monocrystalline Silicon Photovoltaic Microcells as Power Supplies for Transient Biomedical Implants”, *Advanced Energy Materials* **8**, 1703035 (2018).
80. H. Xu#, L. Yin#, C. Liu, **X. Sheng***, N. Zhao*, “Recent Advances in Biointegrated Optoelectronic Devices”, *Advanced Materials* **30**, 1800156 (2018) (**Invited**).
81. Z. Shi, L. Li, Y. Zhao, R. Fu, **X. Sheng***, “Implantable Optoelectronic Devices and Systems for Biomedical Application”, *Chinese Journal of Lasers* **45**, 0207001 (2018) (**Invited**) (**Front Cover**).
史钊, 李丽珠, 赵钰, 付汝兴, **盛兴***, 植入式生物医疗光电子器件与系统, 中国激光, **45**, 0207001 (2018). 主编推荐奖优秀论文
82. R. Nazempour, Q. Zhang, R. Fu, **X. Sheng***, “Biocompatible and Implantable Optical Fibers and Waveguides for Biomedicine”, *Materials* **11**, 1283 (2018) (**Invited**).
83. X. Huang, D. Wang, Z. Yuan, W. Xie, Y. Wu, R. Li, Y. Zhao, D. Luo, L. Cen, B. Chen, H. Wu, H. Xu, **X. Sheng**, M. Zhang, L. Zhao, L. Yin*, “A Fully Biodegradable Battery for Self-Powered Transient Implants”, *Small* **12**, 1800994 (2018) (**Inside Front Cover**).
84. X. Wang, Y. Ma, **X. Sheng**, Y. Wang*, H. Xu*, “Ultrathin Polypyrrole Nanosheets via Space-Confining Synthesis for Efficient Photothermal Therapy in the Second Near-Infrared Window”, *Nano Letters* **18**, 2217 (2018).
85. R. Li, S. Xie, L. Zhang, L. Li, D. Kong, Q. Wang, R. Xin, **X. Sheng**, L. Yin, C. Yu, Z. Yu, X. Wang*,

- L. Gao*, “Soft and transient magnesium plasmonics for environmental and biomedical sensing”, *Nano Research* **11**, 4390 (2018).
86. Y. Yao, K. Lee, **X. Sheng**, N. A. Batara, N. Hong, J. He, L. Xu, M. M. Hussain, H. A. Atwater, N. S. Lewis, R. G. Nuzzo*, J. A. Rogers*, “Porous Nanomaterials for Ultra broadband Omnidirectional Anti-Reflection Surfaces with Applications in High Concentration Photovoltaics”, *Advanced Energy Materials* **7**, 1601992 (2017) (**Frontispiece Cover**).
 87. H. Araki, J. Kim, S. Zhang, A. Banks, K. E. Crawford, **X. Sheng**, P. Gutruf, Y. Shi, R. M. Pielak, J. A. Rogers*, “Materials and Device Designs for an Epidermal UV Colorimetric Dosimeter with Near Field Communication Capabilities”, *Advanced Functional Materials* **27**, 1604465 (2017) (**Back Cover**).
 88. K. Lee, Y. Yao, J. He, B. Fisher, **X. Sheng**, M. Lumb, L. Xu, M. A. Anderson, D. Scheiman, S. Han, Y. Kang, A. Gumus, R. Bahabry, J. W. Lee, U. Paik, N. D. Bronstein, A. P. Alivisatos, M. Meitl, S. Burroughs, M. M. Hussain, J. C. Lee*, R. Nuzzo*, J. A. Rogers*, “Concentrator Photovoltaic Module Architectures With Capabilities for Capture and Conversion of Full Global Solar Radiation”, *Proceedings of the National Academy of Sciences USA*, **113**, E8210–E8218 (2016).
 89. J. Kim, G. A. Salvatore*, H. Araki, A. M. Chiarelli, Z. Xie, A. Banks, **X. Sheng**, Y. Liu, J. W. Lee, K. Jang, S. Y. Heo, K. Cho, H. Luo, B. Zimmerman, J. Kim, L. Yan, X. Feng, S. Xu, M. Fabiani, G. Gratton, Y. Huang, U. Paik*, J. A. Rogers*, “Battery-free, Stretchable Optoelectronic Systems for Wireless Optical Characterization of the Skin”, *Science Advances* **2**, e1600418 (2016).
 90. Y. Yao, L. Xu, **X. Sheng**, N. D. Bronstein, J. A. Rogers, A. P. Alivisatos, R. G. Nuzzo*, “Full solar spectrum conversion via multi-junction architectures and optical concentration”, in *Roadmap on optical energy conversion*, *Journal of Optics* **18**, 073004 (2016) (**Invited**).
 91. X. Guo*, D. Wang, B. Liu, S. Li, **X. Sheng**, “Enhanced light absorption in thin film silicon solar cells with Fourier-series based periodic nanostructures”, *Optics Express* **24**, A408–A413 (2016).
 92. **X. Sheng**#, C. Robert#, S. Wang, G. Pakeltis, B. Corbett*, J. A. Rogers*, “Transfer Printing of Fully Formed Thin-Film Microscale GaAs Lasers on Silicon with a Thermally Conductive Interface Material”, *Laser and Photonics Reviews* **9**, L17–L22 (2015) (**Back Cover**).
 93. **X. Sheng**, M. H. Yun, C. Zhang, A. M. Al-Okaily, M. Masouraki, L. Shen, S. Wang, W. L. Wilson, J. Y. Kim, P. Ferreira, X. Li, E. Yablonovitch, J. A. Rogers*, “Device Architectures for Enhanced Photon Recycling in Thin-Film Multijunction Solar Cells”, *Advanced Energy Materials* **5**, 1400910 (2015) (**Back Cover**).
 94. J. S. Price#, **X. Sheng**#, B. Meulblok, J. A. Rogers*, N. C. Giebink*, “Wide-angle Planar Microtracking for Quasi-static Microcell Concentrating Photovoltaics”, *Nature Communications* **6**, 6223 (2015).
 95. J. S. Price, N. C. Giebink, **X. Sheng**, J. A. Rogers, “Putting CPV on Rooftops”, *Compound Semiconductor Magazine* **21**, 44 (2015) (**Invited**).
 96. **X. Sheng**#, C. A. Bower#, S. Bonafede, J. W. Wilson, B. Fisher, M. Meitl, H. Yuen, S. Wang, L. Shen, A. R. Banks, C. J. Corcoran, R. G. Nuzzo, S. Burroughs*, J. A. Rogers*, “Printing-based Assembly of Quadruple Junction, Four-terminal Microscale Solar Cells and Their Use in High-efficiency Modules”, *Nature Materials* **13**, 593–598 (2014).
 97. **X. Sheng**#, C. Yu#, V. Malyarchuk, Y. Lee, S. Kim, T. Kim, L. Shen, C. Horng, J. Lutz, N. C. Giebink, J. Park, J. A. Rogers*, “Silicon based visible-blind ultraviolet detection and imaging using down-shifting luminophores”, *Advanced Optical Materials* **2**, 313 (2014) (**Frontispiece Cover**).
 98. **X. Sheng***, L. Z. Broderick, L. C. Kimerling, “Photonic crystal structures for light trapping in thin-film Si solar cells: modeling, process and optimizations”, *Optics Communications* **314**, 41 (2014) (**Invited**).
 99. H. Ning, N. A. Krueger, **X. Sheng**, H. Keum, C. Zhang, K. D. Choquette, X. Li, S. Kim, J. A. Rogers, P. V. Braun*, “Transfer printing of tunable porous silicon microcavities with embedded emitters”,

ACS Photonics **1**, 1144–1150 (2014).

100. Y. Shen, Y. Jia, **X. Sheng**, L. Shen, J. A. Rogers, N. C. Giebink*, “Nonimaging optical gain in luminescent concentration through photonic control of emission etendue”, *ACS Photonics* **1**, 746–753 (2014).
101. Y. Zou, **X. Sheng**, K. Xia, H. Fu, J. Hu*, “Parasitic loss suppression in photonic and plasmonic photovoltaic light trapping structures”, *Optics Express* **22**, A1197–A1202 (2014).
102. **X. Sheng**#, L. Shen#, T. Kim, L. Li, X. Wang, R. Dowdy, P. Froeter, K. Shigeta, X. Li, R.G. Nuzzo, N. C. Giebink*, J. A. Rogers*, “Doubling the power output of bifacial thin-film GaAs solar cells by embedding them in luminescent waveguides”, *Advanced Energy Materials* **3**, 991–996 (2013) (**Front Cover**).
103. **X. Sheng**#, C. J. Corcoran#, J. He, L. Shen, S. Kim, J. Park, R. G. Nuzzo*, J. A. Rogers*, “Enhanced ultraviolet responses in thin-film InGaP solar cells by down-shifting”, *Physical Chemistry Chemical Physics* **15**, 20434–20437 (2013).
104. **X. Sheng***, J. Hu, J. Michel, L. C. Kimerling, “Light trapping limits in plasmonic solar cells: an analytical investigation”, *Optics Express* **20**, A496–A501 (2012).
105. **X. Sheng***, S. G. Johnson, L. Z. Broderick, J. Michel, L. C. Kimerling, “Integrated photonic structures for light trapping in thin-film Si solar cells”, *Applied Physics Letters* **100**, 111110 (2012).
106. **X. Sheng**, J. Liu, I. Kozinsky, A. M. Agawal, J. Michel*, L. C. Kimerling, “Design and non-lithographic fabrication of light trapping structures for thin film silicon solar cells”, *Advanced Materials* **23**, 843–847 (2011).
107. **X. Sheng***, S. G. Johnson, J. Michel, L. C. Kimerling, “Optimization-based design of surface textures for thin-film Si solar cells”, *Optics Express* **19**, A841–A850 (2011).
108. **X. Sheng***, L. Z. Broderick, J. Hu, L. Yang, A. Eshed, E. A. Fitzgerald, J. Michel, L. C. Kimerling, “Design and fabrication of high-index-contrast self-assembled texture for light extraction enhancement in LEDs”, *Optics Express* **19**, A701–A709 (2011).
109. **X. Sheng***, J. Liu, N. Coronel, A. M. Agawal, J. Michel, L. C. Kimerling, “Integration of self-assembled porous alumina and distributed bragg reflector for light trapping in Si photovoltaic devices”, *IEEE Photonics Technology Letters* **22**, 1394–1396 (2010).
110. X. Zhou, Z. Li, Y. Wang, **X. Sheng**, Z. Zhang*, “Photoluminescence of amorphous niobium oxide films synthesized by solid-state reaction”, *Thin Solid Films* **516**, 4213–4216 (2008).
111. G. Sheng, Z. Li*, **X. Sheng**, Y. Hu, Z. Zhang, “Microcosmic behavior research of palladium membrane irradiated by helium ions”, *Atomic Energy Science Technology* **41**, 418 (2007)
盛国福, 李正操*, **盛兴**, 胡殷, 张政军, 氦离子辐照下金属钯薄膜的微观行为研究, 原子能科学与技术, **41**, 418 (2007).
112. Y. Wang, Z. Li, **X. Sheng**, Z. Zhang*, “Synthesis and optical properties of V₂O₅ nanorods”, *Journal of Chemical Physics* **126**, 164701 (2007).

Book Chapters:

1. C. Liu, **X. Sheng**, “Transfer-Printed Devices for Biomedical Applications”, in *Transfer Printing Technologies and Applications* ed. by C. Cao and Y. Sun, Elsevier (2024).
2. H. Ding, **X. Sheng**, “Thin-Film III-V Single Junction and Multijunction Solar Cells and Their Integration onto Heterogeneous Substrates”, in *Inorganic Flexible Optoelectronics: Materials and Applications* ed. by Z. Ma and D. Liu, Wiley-VCH (2019).
3. **X. Sheng**, S. Wang, L. Yin, “Flexible, Stretchable and Biodegradable Thin-Film Silicon Photovoltaics”, in *Advances in Silicon Solar Cells* ed. by S. J. Ikhmayies, Springer-Verlag (2018).

4. L. Yin, **X. Sheng**, “Nonconventional Biosensors Based on Nanomembrane Materials”, in *Nanobiomaterials: Classification, Fabrication and Biomedical Applications* ed. by X. Wang, M. Ramalingam, X. Kong and L. Zhao, Wiley-VCH (2018).
5. **X. Sheng**, *Thin-film Silicon Solar Cells: Photonic Design, Process and Fundamentals*, LAMBERT Academic Publishing (2012).

Patents:

1. **盛兴**, 王华春, 基于光电调控的三维生物支架, ZL 2023 1 0140332.9
2. **盛兴**, 孔德颖, 尹斓, 一种钙钛矿单晶薄膜的制备方法及其应用, ZL 2022 1 0657214.0
3. **盛兴**, 赵钰, 谢杨, 植入式探针及其制备方法、控制电路及控制系统, ZL2021 1 0376512.8
4. **盛兴**, 赵钰, 谢杨, 植入式探针、控制电路及控制系统, ZL2021 2 0717664.5
5. **盛兴**, 刘长波, 李丽珠, 赵钰, 蔡雪, 谢杨, 王强, 多功能植入式探针及其制备方法, ZL2020 1 0874368.6
6. **盛兴**, 丁贺, 史钊, 上转换器件和材料及其制造方法, ZL2017 1 1212171.0
7. **X. Sheng**, H. Ding, Z. Shi, “Optoelectronic Upconversion Devices”, PCT WO2019100380.
8. J. A. Rogers, **X. Sheng**, C. A. Bower, M. Meitl, S. Burroughs, “Printing-based multi-junction, multi-terminal photovoltaic devices”, US20150207012 / WO2015109242.
9. A. Agarwal, B. Albert, L. Z. Broderick, J. Cheng, J. Hu, L. C. Kimerling, J. Liu, J. Michel, **X. Sheng**, “Methods and apparatus for concentration photovoltaics”, US20140090686 / WO2013056139.
10. **X. Sheng**, J. Liu, J. Michel, A. M. Agarwal, L. C. Kimerling, “Pseudo-periodic structure for use in thin film solar cells”, US20100307579 / WO2010141145.

Services

Internal at Tsinghua:

- *Panelist in postdoc searching committee*
- *Panelist in graduate admission committee*
- *Panelist in undergraduate admission committee*
- *Panelist in graduate thesis committee*
- *Freshmen Mentor*
- *Supervising undergraduate students supported by the Student Research Training (SRT) program*

External:

- *Society Membership*
 - *Optica Fellow*
 - *IEEE senior member*
 - *SPIE life member*
- *Journal Editor*
 - *Optical Materials Express*, Associate Editor, 2017–2023.
 - *Fundamental Research*, Special issue “Multimodal Technologies for Neural Modulation and Sensing”, Lead Editor, 2023.
 - *IEEE Journal of Selected Topics in Quantum Electronics*, Special issue “Flexible Optoelectronics”,

Guest Editor, 2023.

- *Frontiers in Nanotechnology*, Feature issue “New Technologies for Large-Scale Recording and Modulation in the Brain”, Guest Editor, 2021.
- *Optical Materials Express*, Feature issue “Bio-inspired and Bio-integrated Photonic Materials and Devices”, Lead Editor, 2019.
- *Board Member*
 - *Chinese Association of Automation*
 - *Chinese NeuroScience Society*
 - *Chinese Society of Biomedical Engineering*
- *Conference Organizer for multiple domestic and international conferences*
 - 2024 Nature Conference on Transformative Technologies for Neuroengineering, Shenzhen, China. Co-Organizer.
 - 2023 Optica Advanced Photonics Congress, Solar Energy and Light-Emitting Devices (SOLET) Topical Meeting, Busan, Korea. Subcommittee.
 - 2020 CIMTEC 9th Forum on New Materials, Montecatini Terme, Italy. International Advisory Board Member.
 - 2019 IEEE-EMBS 16th International Conference on Wearable and Implantable Body Sensor Networks (BSN), Chicago, IL, USA. Technical Program Committee.
 - 2019 MRS spring meeting, Phoenix, AZ, USA. Symposium Organizer.
 - 2017 OSA IPR meeting, New Orleans, LA, USA. Subcommittee.
 - 2016 MRS fall meeting, Boston, MA, USA. Symposium Organizer.
 - 2016 MRS spring meeting, Phoenix, AZ, USA. Symposium Organizer.
- *Reviewer for multiple international journals*
- *Proposal Reviewer for NSFC, and multiple international funding agencies*
- *Co-president, MIT Chinese Association of Science and Technology, 2010.*
- *Scientific consultant for several high-tech start-up companies*

Awards and Honors

International

- Optica Fellow, 2024
- Rising Stars of Light (Finalists), by Light: Science & Applications, 2023
- Young Scientist Award, Photonics & Electromagnetics Research Symposium (PIERS), 2018
- Young Scientist Award, Microsystems & Nanoengineering Summit (MINE), 2018
- Best Poster Award (2nd prize) in Nature Conference on Flexible Electronics, Nanjing, 2016
- Gordon Engineering Leadership Teaching Assistantship, MIT, 2011
- Energy Initiative Seed Fund Award, MIT, 2010
- Best Poster Award (runner-up) in the 35th IEEE Photovoltaic Specialists Conference, 2010
- Energy Initiative Martin Fellowship, MIT, 2010
- DuPont-MIT Alliance Fellowship, 2007
- MIT Presidential Fellowship, 2007

Domestic (in Chinese)

- 2024, 国家自然科学基金委交叉科学部杰出青年科学基金
- 2024, 《中国激光》“生物医学光子学-最美封面论文”
- 2022, 清华大学, 电子工程系, 郑君里教书育人优秀教师奖
- 2022, 清华大学, 电子工程系, 周炳琨学者奖
- 2022, 中国生物医学工程大会, 青年优秀论文报告
- 2021, 中国材料研究学会, 科学技术一等奖(基础研究类)(排名第三)
- 2019, 《中国激光》主编推荐奖优秀论文
- 2018, “中国新锐科技卓越影响奖”
- 2014, 国家海外人才计划青年项目
- 2007, 清华大学, 优良毕业生
- 2006, 清华大学, 杜邦学生奖学金
- 2005, 清华大学, 三星学生奖学金
- 2004, 清华大学, 伍占德学生奖学金
- 2003, 清华大学, 新生奖学金